

Erdős Problem #1132

EP-1132

L1 - TRACTABLE

PARTIALLY SOLVED

Combinatorics

ORIGINAL PROBLEM STATEMENT

For $x_1, \dots, x_n \in [-1, 1]$ let $l_k(x) = \frac{\prod_{i \neq k} (x - x_i)}{\prod_{i \neq k} (x_k - x_i)}$, which are such that $l_k(x_k) = 1$ and $l_k(x_i) = 0$ for $i \neq k$.

Let $x_1, x_2, \dots \in [-1, 1]$ be an infinite sequence, and let $L_n(x) = \sum_{1 \leq k \leq n} |l_k(x)|$, where each $l_k(x)$ is defined above with respect to $x_1, \dots, x_n$.

Must there exist $x \in (-1, 1)$ such that $L_n(x) > \frac{2}{\pi} \log n - O(1)$ for infinitely many n ?

Is it true that $\limsup_{n \rightarrow \infty} \frac{L_n(x)}{\log n} \geq \frac{2}{\pi}$ for almost all $x \in (-1, 1)$?

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Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>